

5(a)

Using $v = f\lambda$,

$$\begin{aligned}\lambda &= \frac{v}{f} \\ &= \frac{340}{1700} \\ &= 0.200 \text{ m}\end{aligned}$$

The inter-nodal distance is half a wavelength, i.e. 0.100 m.

Particles of the waves at the node do not oscillate whereas particles at the antinodes oscillate with the maximum amplitude. Hence, the resultant stationary wave has maximum intensity at the antinode positions.

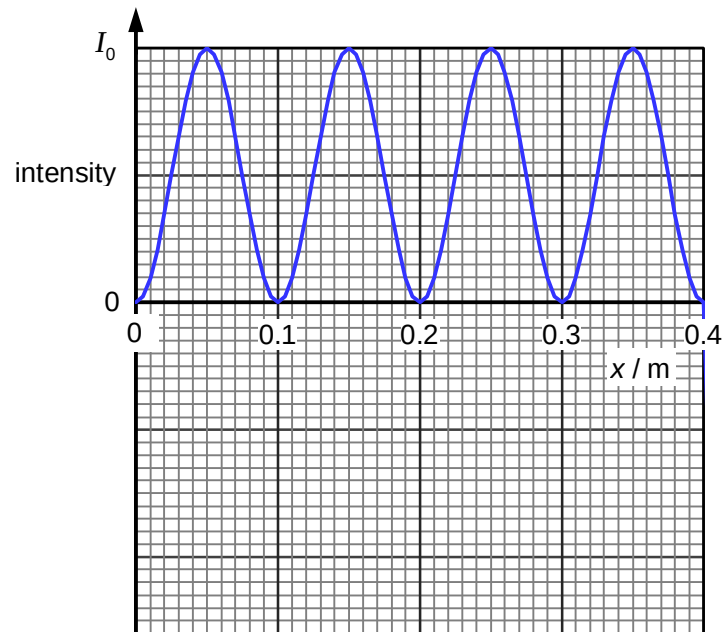


Fig. 5.1

Note: Important to realise that the intensity could not have negative values, and the graph must display a clear $\sin^2$ graph.

No.	Solution
5(b)	Since $T = \frac{1}{f}$, $T = \frac{1}{250}$ = 0.0040 s = 4.0 ms Since the amplitude of the wave at X varies sinusoidally with time, the variation with time of the change in air pressure due to the wave passing X is sinusoidal. Fig. 5.2 <i>Note: Both sine or cosine graph with the correct time period is acceptable.</i>

No.	Solution
5(c)	The figure shows a graph of magnetic flux density B on the vertical axis against radius r on the horizontal axis. The vertical axis has major grid lines at 0, $0.33B_0$, $0.5B_0$, and B_0. The horizontal axis has major grid lines at 0, r, $2r$, and $3r$. A blue curve is plotted, starting at the point (r, B_0) and decreasing as the radius increases. The curve passes through the point $(2r, 0.5B_0)$ and ends at the point $(3r, 0.33B_0)$. The area under the curve is shaded with a light blue grid. Fig. 5.3 Use $B = \frac{\mu_0 NI}{2r}$, for constant current in the coil and constant number of turns, $B \propto \frac{1}{r}$ <i>Note: The instruction to cover a range of r to $3r$ has to be observed strictly.</i>
6(a)(i)	Potential difference across the lamp